WHITE PAPER

Applying Quantum Mechanics to Classical Computing

An introduction to computation, storage, memory and data movement in the NSATS Quantum portfolio

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Abstract

NSATS licences enable classical devices to perform operations based on quantum mechanics, at room temperature. This paper examines reversibility as a connection between quantum mechanics and classical computation, then describes its place within the NSATS Quantum portfolio. The discussion covers chip design, storage, memory, internal buses, networking and security, with licensees responsible for building the implementations. The licensed technology supports faster processing and data transfer, increased effective storage capacity, greater usable memory capacity and lower energy use. For infrastructure decisions, the central question is how those implementations serve a defined workload. Evaluation considers task completion, retained information, data movement, energy use and security across the system. Scientific background and a worked example explain information preservation; the application sections distinguish effective storage capacity, processing rate and working memory. An illustrative workflow, decision table and short protocol show how an organisation can structure its evaluation.

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Computing as a complete task

A computing task includes calculation, memory access, storage and communication. A processor waiting for an input cannot complete the operation that requires it. A calculated result must also reach the application or person who will use it. An infrastructure assessment therefore needs to account for the path from stored input to usable output.

Consider an engineering team analysing a collection of measurements. Its system reads the stored measurements, supplies them to an analysis program, performs calculations and retains the output. If another computer needs the results, the system also prepares and transfers them. In this illustrative workflow, the task ends when the intended user or application can access the analysis output.

General scientific background

Reversibility provides a connection between quantum mechanics and classical computation through the preservation of information.

An isolated quantum system has an ideal form of evolution called unitary evolution. This transformation is reversible: its mathematical inverse returns the initial state. Measurement and interaction with an environment require a description beyond isolated unitary evolution. The useful conceptual connection to computing is information preservation: a reversible transformation keeps the distinctions needed to identify a preceding state. [1]

This scientific background concerns information preservation; the NSATS technology is described separately below.

Applying the technology in classical devices

NSATS's Quantum technology applies computational relationships drawn from quantum mechanics in classical electronic circuits. It is developed and available under licence, and operates without super-cooling. Its hardware basis is specially designed classical chipsets, not off-the-shelf third-party hardware.

Licensees develop chipsets, hard drives, memory (RAM), networking equipment, computer buses and other technical infrastructure incorporating the licensed technology.

NSATS sells technology licences, not chips. Its technology helps licensees manufacture their own chips.

NSATS/DS062026/V2.000

NSATS licenses its intellectual property: patents, designs and copyrighted software. Where the licensed IP takes the form of software, NSATS supplies it under the licence, with its documentation and updates. Each licensee designs and builds its own solution.

The division of functions between software and hardware depends on the technology concerned and the licensee's design.

Reversible Computing

Reversible computing preserves the information needed to trace a computation back to its input. If different inputs produce the same result, that result alone does not identify the original input. Retaining the relevant supporting information preserves that distinction.

A worked example

Adding three and seven gives ten. Adding four and six also gives ten. If only the total is kept, it does not identify the original pair. Suppose the operation retains both the total, ten, and the first input, three. Subtracting three from ten returns seven, so the pair can be identified. The retained output is the total together with the supporting input. This arithmetic example illustrates information preservation; it does not describe NSATS circuitry or compression.

Information erasure and energy

An irreversible erasure discards the distinction between possible prior information states. Landauer's analysis connects such logical irreversibility with heat generation in a physical computing device. Preserving information through reversible operations avoids that erasure step within the operation. This explains why reversibility is relevant to the study of energy-efficient computation. Practical energy use also depends on the physical implementation, memory and data movement; logical reversibility alone does not determine a system's energy consumption. [2]

Retaining information also involves computational resources. Bennett's analysis of reversible computation examines the processing time and storage space required to carry out an ordinary computation reversibly while saving its input. The trade-off concerns the resources used to preserve and manage that information. [3] The implementation assessment in this paper therefore includes the work of retaining and accessing supporting information.

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NSATS's processor-assisted, semi-reversible computing

NSATS has achieved reversible computing at component level: its swap gates retain input and output information and work reversibly in a limited form. An external processor uses the retained information to regenerate earlier inputs. Full reversibility across the complete system is the subject of continuing research. Earlier information can then be used in subsequent calculations. White Paper 2, *Reversible Computing: Preserving Information Through Computation*, examines this arrangement, which NSATS describes as processor-assisted, semi-reversible computing.

Chip Design

A logic gate defines an operation on information. A chip combines gates with connections that carry inputs and results between them.

NSATS's portfolio includes several quantum-inspired gate designs that improve the efficiency of operations carried out in chips and increase chipset speed. Licensees combine the licensed designs within their implementations. The designs can be incorporated into programmable hardware or custom chips. Programmable hardware allows selected electronic functions to be configured; a custom chip places functions within a dedicated design.

The licensee determines which functions belong in hardware and which run in software, including how each receives its working data.

Processors and Internal Buses

A processor carries out a program's operations. It receives inputs from memory or other components and sends results through the device's internal connections.

A bus is a communication path carrying signals between components. The quantum-classical bus carries electrical signals with more than two states, beyond binary 0 and 1. The bus enables faster processing, data transfer and decision making within a quantum-classical electronic device.

Temporal Compression and storage

Temporal compression increases effective storage capacity, allowing more data to be stored within the same physical storage resource. Physical capacity describes the underlying storage resource; effective capacity describes the amount of information the complete implementation stores and retrieves. The drive's physical size stays unchanged.

NSATS/DS062026/V2.000

For existing drives and RAM, the licensee adds a specially designed classical chipset or controller built with the licensed technology. It encodes classical data using quantum-mechanical logic and manages intermediate storage and retrieval operations. The drive or RAM itself remains ordinary classical hardware. This arrangement applies to hard drives, solid-state drives, USB flash drives, optical storage and other storage types, as well as the working-memory use described in the next section.

NSATS's processor-assisted, semi-reversible computing contributes to compression, so the drive holds more data. The resulting effective capacity depends on the storage device, how the licensee uses the technology and where the licensed gates, buses and computational technologies sit within the design.

A temporal drive is a storage implementation that applies NSATS's temporal compression through the added hardware described above. The term connects its storage capability with the processing time involved in compressing and decompressing data and in storing and retrieving it. The following description concerns how that drive is rated.

Effective capacity depends on processing time, not only physical size. A temporal drive is better described by how much data it can process and store per second than by a fixed capacity.

A storage assessment needs both the processing rate and the amount retained: how quickly the system stores and retrieves data, and how much information it holds for retrieval.

Memory and RAM

RAM holds the working information a program needs during execution; retained storage holds data for later use.

Through the added hardware described under Temporal Compression and storage, existing classical RAM holds classical data encoded using quantum-mechanical logic, so computer buses can send and retrieve it faster. NSATS's memory technology supports greater usable memory capacity and faster access to working information.

Access latency is the delay before requested data is available; bandwidth is the rate at which data is transferred. A program's sequence of accesses affects how these characteristics influence its work. Memory-controller designers have a role alongside RAM manufacturers because the controller coordinates access and connects memory activity with processor operation.

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Quantum-Classical Networking

Networking extends data movement between systems. Licensee implementations built with NSATS's licensed technology use quantum-classical network cards at both the sending and receiving endpoints to enable high-speed data transfer. Each card contains a chip that handles processing associated with transmission or receipt, while the data passes through ordinary routers in between.

Transfer assessment includes preparation at the source, movement through the network and reconstruction at the destination. Completion is the point at which the receiving application can use the information.

Licensee implementations can serve transfers over wired, optical and wireless links, including long-distance connections. Compatibility concerns the endpoint systems and the ordinary equipment along the network path.

Post-Quantum Security

Post-quantum security addresses the future threat that quantum computing poses to widely used public-key cryptographic methods. Protection must also address classical threats and the operation of the surrounding system. NIST's introductory report explains these goals and the importance of interoperability. [4]

NSATS's post-quantum security uses classical computing methods inspired by quantum mechanics. For browsers and the servers they connect to, NSATS licenses the security as software under standard software licence terms. The software runs in the browser and on the server. For larger systems, licensees build the security into their own chips.

The information to protect includes data in transit, while it travels between systems, and data at rest, while it is stored. Within chip-based systems, the security scope also includes information during processing, in memory and along internal data paths.

Applications

For storage and memory manufacturers, the technology concerns effective capacity, access to data and controller integration. For networking manufacturers, it concerns processing at both ends of a transfer and compatibility along the path. Product qualification needs to reflect the complete implementation that the manufacturer builds.

NSATS/DS062026/V2.000

Applications include high-performance computing and new computing platforms using processor-assisted, semi-reversible computing. Scientific simulations and engineering analysis bring calculations, memory access and data movement into a single workload. Browser and application providers address protection of information across the client, server and storage involved, using the security scope described above.

Data centre and hosting providers consider storage resources, working memory, processing, power and heat within their intended systems. With the licensed technology, fewer hard drives and less memory are needed to store data, existing memory processes existing binary data faster, and the chipsets run with lower heat than conventional computing. The extent depends on the deployed design and workload. Chipset heat, server power and facility cooling describe different parts of the infrastructure and need to be measured separately.

Assessing an implementation

Illustrative assessment: engineering measurements

Returning to the illustrative engineering workflow introduced above, suppose the team needs to analyse a selected collection of measurements, retain the results and make them available to an application on another computer. The team assesses a licensee-built implementation against its existing configuration. The assessment follows that task from the stored input to the receiving application.

At the input stage, the evaluator checks that the selected measurements are retrieved intact and records the work needed to supply them to the analysis program. During calculation, the evaluator records working-memory use, access delays and processing time, including the work of retaining and accessing supporting information. These observations help distinguish time spent calculating from time spent waiting for data.

At the output stage, the evaluator checks the calculated result against the expected result defined for the task, then verifies that the stored output can be retrieved intact. If the output is transferred, the observation continues through preparation at the source, the network path and reconstruction at the destination. The task is complete when the receiving application has the required information and can use it. A record of calculation time alone would leave the storage and transfer work unaccounted for.

NSATS/DS062026/V2.000

Decisions and evidence

The table identifies the relevant part of the implementation and the evidence to examine before deciding whether it meets the workload's requirements. The stated outcome directions are those described in this paper, which the organisation's assessment tests against its baseline for its own workload; the security row specifies protection scope only.

Workload constraint or requirement	Implementation to examine	Stated outcome direction to test	Evidence for the decision
Completing the task within the required time	Processor, internal buses, memory and storage	Faster processing, data transfer and decision making through the bus; gate designs that increase chipset speed.	Time from input preparation to usable output, with calculation, data access and management of retained information identified separately.
Retaining the required information and retrieving it intact	Storage device and added chipset or controller	Increased effective storage capacity; fewer hard drives needed to store data.	Amount retained, integrity of retrieved data, and the processing needed for storage and retrieval. Record the physical resource separately from effective capacity.
Supplying working data during execution	RAM, added controller and processor connection	RAM data sent and retrieved faster; greater usable memory capacity.	Working-memory use, access latency and bandwidth for the selected workload, considered alongside task completion.
Delivering usable information to another system	Both endpoint cards and the intervening network path	High-speed data transfer between the endpoint cards.	Completion at the receiving application, accounting for preparation and reconstruction, together with integrity and compatibility checks across the path.
Operating within power and cooling requirements	The components involved in the task, the server and facility cooling	Lower energy use; chipsets running with lower heat than conventional computing.	Energy consumed by the task and separate records for chipset heat, server power and facility cooling, with operating conditions documented.
Protecting information	The applicable software or chip-based security implementation and its surrounding system	Protection scope: data in transit and at rest; for chip-based systems, information during processing, in memory and along internal data paths.	Defined threats, key-access responsibilities and the scope and findings of any independent security assessment.

Assessment protocol

An organisation can structure its assessment as follows. It sets its own acceptance criteria according to the intended use.

1. **Define the task and acceptance criteria.** Specify the input, expected output, completion point and requirements for retention, access, energy use, integrity and security. Assign responsibility for reviewing the evidence and making the decision before the assessment begins.
2. **Document the configuration and data.** Record the hardware and software involved, the added controller or chipset where applicable, both endpoints for networking, and the operating conditions. Use the existing configuration, or another defined reference configuration where none exists, as the baseline, holding the task, data and operating conditions constant across the baseline and the licensee-built implementation. Select data representative of the application, including already compressed or encrypted files where relevant.
3. **Observe the complete task.** Use the observations described in the illustrative scenario. Repeat observations with the task, data and operating conditions held constant to examine consistency, recording failures and any configuration changes as well as successful completion.
4. **Interpret the results.** Assess the evidence against the baseline as well as the agreed criteria, identifying which requirements were met, which were not met and which remain unresolved. Keep conclusions tied to the configuration and data examined. Where evidence is incomplete, specify what further assessment is needed before making the implementation decision.

Conclusion

The scientific discussion connects reversibility with information preservation and the energy implications of erasure. NSATS applies quantum-mechanical concepts in licensed technology for classical computing, storage, memory, communication and security. Comparing a licensee-built implementation with an existing configuration for a defined workload connects the stated outcomes to the organisation's decision. The organisation can use that evaluation to frame a licensing discussion around its intended implementation.

NSATS/DS062026/V2.000

Scientific background references

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NSATS/DS062026/V2.000